

Notes: Hong and Stein (JF, 1999)

A Unified Theory of Underreaction, Momentum Trading, and Overreaction in Asset Markets

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1 Big picture

- **Question.** Can one unified behavioral model explain both **short-run momentum** / **continuation** and **long-run reversal** / **overreaction** in asset prices?
- **Setting.** The paper builds a simple **infinite-horizon asset-pricing model** with heterogeneous boundedly rational traders. The goal is not to fit one specific dataset, but to explain a broad set of stylized facts in stock returns.
- **Core challenge.** Standard rational models struggle to generate predictable returns at both short and long horizons. Existing behavioral models in the late 1990s mostly relied on a representative investor with cognitive biases. Hong and Stein instead want a mechanism driven by **interaction across trader types**.
- **Main theoretical idea.** The market contains two groups:
 - **Newswatchers:** they observe private information about fundamentals, but do *not* infer information from prices.
 - **Momentum traders:** they condition on past price changes, but only through **simple univariate forecasting rules**.
- **Primitive source of underreaction.** Private information diffuses gradually across the population of newswatchers. Since they do not learn from prices, prices respond only gradually to new information.
- **Primitive source of overreaction.** Momentum traders try to exploit this underreaction by trend-chasing. But because they cannot distinguish an **early** price increase caused by fresh news from a **late** price increase caused by earlier momentum trades, they can push prices beyond fundamentals.
- **Main theoretical result.** Underreaction and overreaction come from the **same shock**: gradually diffusing information about fundamentals. The model does *not* need separate sentiment shocks or liquidity shocks.
- **Why it matters.** The paper gives a parsimonious explanation for two apparently different anomalies:
 - positive return autocorrelation at horizons of roughly 3–12 months,
 - negative return autocorrelation / reversal at longer horizons.
- **Core mechanism in plain English.** Newswatchers create a slow drift toward fundamental value. Momentum traders accelerate that drift, but because they use simple strategies, they create a **momentum externality**: early trend followers make money by buying ahead of later trend followers, and late trend followers get trapped after prices have already overshot.
- **Main model implications.**
 - There is **underreaction** when only newswatchers trade.
 - With momentum traders, there is **always overreaction** in the baseline covariance-stationary equilibrium.

- More sophisticated arbitrage — contrarians or finite-risk-tolerance “smart money” — attenuates the hump shape, but does not eliminate it unless arbitrage becomes effectively infinite and fully rational.
- **Main empirical predictions.**
 - Momentum strategies should work best where information diffuses most slowly, such as **small** and **low-analyst-coverage** stocks.
 - The same stocks should also be the most **reversal-prone** at long horizons.
 - Private-news shocks should be more likely than public-news shocks to produce a **hump-shaped impulse response**.
 - The horizon at which autocorrelations switch sign should depend on **momentum traders’ holding horizons** and hence plausibly on trading costs.
- **Economic takeaway.** Once some traders underreact to slowly diffusing information, even nearly rational arbitrage can produce overreaction if arbitrageurs rely on simple trend-following rules. In this sense, **underreaction sows the seeds of overreaction**.

2 Data and measurement (key objects)

2.1 The stylized facts the model is designed to explain

- **Short-run continuation.** The paper starts from the evidence that asset returns display positive serial correlation at horizons around **3 to 12 months**. This includes both unconditional momentum in past returns and post-event drift after public announcements.
- **Examples of conditional continuation.** The paper highlights post-event drift after:
 - earnings announcements,
 - stock issues and repurchases,
 - dividend initiations and omissions,
 - analyst recommendations.
- **Long-run reversal.** At longer horizons, the paper takes as given the evidence that past winners tend to underperform and past losers tend to outperform — the DeBondt–Thaler style overreaction pattern — together with related fundamental-reversion evidence based on variables such as book-to-market and cash-flow-to-price ratios.
- **Why this matters.** The model is built to explain these two facts *jointly*, not one at a time.

2.2 Asset, fundamentals, and shocks

- **Single risky asset.** The economy has one risky asset that pays a terminal liquidating dividend at date T .
- **Dividend process.** The final dividend is

$$D_T = D_0 + \sum_{j=0}^{\infty} \varepsilon_j,$$

where the shocks ε_j are independent, mean-zero normal innovations.

- **Subinnovations.** Each dividend innovation can be decomposed into z independent subinnovations. This is how the paper models gradual information diffusion.
- **Limiting horizon.** The analysis takes the limit as $T \rightarrow \infty$, which allows the authors to focus on **steady-state trading strategies**.

2.3 Trader types and information sets

- **News watchers.**
 - The population is divided into z equal-sized groups.
 - At time t , each group observes only one fraction of a future dividend innovation.
 - Information about a given shock rotates across groups over time, so it gradually becomes public after z periods.
 - News watchers use the private signals they directly observe, but **ignore current and past prices** when forecasting fundamentals.
 - They also behave as if they **buy and hold until liquidation**, which is a deliberate simplifying shortcut.
- **Momentum traders.**
 - A new generation enters every period.
 - Each generation holds its position for j periods.
 - Momentum traders cannot see the private signals ε directly.
 - They forecast future price changes using only past price changes, in the baseline with a **single lag of price momentum**.
- **Possible extensions.**
 - contrarians who bet against older price changes,
 - “smart money” traders who use the full information set and all past prices, but still have finite risk tolerance.

2.4 Timing and the meaning of the key parameters

- **z : information diffusion parameter.** Higher z means slower information flow. A shock takes more periods to become widely known.
- **j : momentum traders’ horizon.** This is how long a generation of momentum traders holds its position.
- **γ : risk tolerance of momentum traders or other arbitrageurs.** Higher risk tolerance makes arbitrage more aggressive.
- **Q : exogenous net supply of the asset.**

- **ϕ : momentum-trading elasticity.** This is the strength with which past price changes translate into current order flow.
- **w : fraction of arbitrage capital allocated to momentum rather than contrarian trading in the extension with multiple simple strategies.**
- **Interpretation of the state variables.** The model is really about how the market responds to a single dose of gradually diffusing news, and how that news interacts with the strategic positions of arbitrageurs over time.

3 Models and empirical specifications (all equations + interpretation)

3.1 Dividend process and rational expectations benchmark

The final payoff of the risky asset is

$$D_T = D_0 + \sum_{j=0}^{\infty} \varepsilon_j.$$

The paper takes the limit as $T \rightarrow \infty$ and defines a rational-expectations benchmark price:

$$P_t^* = D_{t+z-1} - Q.$$

Interpretation. Under full rational expectations and fully revealing prices, the price would immediately aggregate all currently available private information. The benchmark therefore jumps directly to the correct conditional expectation of long-run value.

3.2 Price formation with newswatchers only

When only newswatchers are active, the price at time t is

$$P_t = D_t + \frac{(z-1)\varepsilon_{t+1} + (z-2)\varepsilon_{t+2} + \cdots + \varepsilon_{t+z-1}}{z} - Q.$$

Interpretation.

- The shock enters price only gradually because each period only part of the population has seen it.
- The weights fall linearly because more distant innovations are less widely known.
- Prices therefore **underreact** to newly arriving private information.
- Importantly, this newswatcher-only economy does **not** generate overreaction: prices drift toward value, but do not overshoot.

3.3 Momentum-trader order flow

In the baseline, momentum traders forecast future price changes using only the most recent price change. Their order flow is:

$$F_t = A + \phi \Delta P_{t-1},$$

where $\Delta P_{t-1} = P_{t-1} - P_{t-2}$.

If there are j overlapping generations of momentum traders active in the market, aggregate supply absorbed by newswatchers is

$$S_t = Q - jA - \sum_{i=1}^j \phi \Delta P_{t-i}.$$

Interpretation.

- Positive recent returns induce current demand from momentum traders.
- Since positions stay open for j periods, today's price depends on several past rounds of momentum order flow.
- This overlap of generations is what creates internal propagation of price pressure.

3.4 Equilibrium price with momentum traders

Replacing fixed supply Q by the effective supply absorbed by newswatchers, the equilibrium price becomes

$$P_t = D_t + \frac{(z-1)\varepsilon_{t+1} + (z-2)\varepsilon_{t+2} + \cdots + \varepsilon_{t+z-1}}{z} - Q + jA + \sum_{i=1}^j \phi \Delta P_{t-i}.$$

Interpretation.

- The first block is the slowly diffusing fundamentals part.
- The last term is the endogenous momentum-amplification term.
- Price today depends on past price changes, which themselves were partly caused by earlier momentum orders. This is the source of hump-shaped dynamics.

3.5 Momentum-trader optimization and the fixed point

Momentum traders choose ϕ optimally. In equilibrium,

$$\phi = \gamma \frac{\text{Cov}(P_{t+j} - P_t, \Delta P_{t-1})}{\text{Var}(\Delta P) \text{Var}_M(P_{t+j} - P_t)}.$$

Interpretation.

- The numerator measures how informative the latest price change is about the return over the trader's holding horizon.

- The denominator scales by the volatility of the signal and the conditional risk of the position.
- Equilibrium is a **fixed point**: ϕ affects price dynamics, which affect forecastability, which in turn determine the optimal ϕ .

The paper focuses on **covariance-stationary equilibria**. A key lemma is:

Lemma 1. In any covariance-stationary equilibrium, $\phi > 0$.

Interpretation. Trend-chasing is not imposed by assumption once the model is set up this way; it follows from the fact that lagged price increases are informative about future returns when news diffuses slowly.

3.6 Proposition 1: overreaction in the impulse response

Suppose a positive one-unit shock begins to diffuse at time t . Then:

Proposition 1.

1. There is always **overreaction**: the cumulative impulse response of prices peaks above one.
2. If $j \geq z - 1$, the cumulative impulse response peaks at $t + j$ and then declines, eventually converging back to one.
3. If $j < z - 1$, the peak occurs no earlier than $t + j$, and the impulse response still eventually converges to one.

Interpretation.

- Newswatchers push price up gradually as information diffuses.
- Momentum traders add more buying on top of this drift.
- Since early momentum traders cause price increases that later momentum traders partly misread as fresh news, prices go too far.
- Eventually, as old positions unwind and the news is fully absorbed, prices come back toward fundamentals.

3.7 Proposition 2: negative autocorrelation must eventually show up

With risk-neutral momentum traders, if returns are positively correlated at short horizons, then they must be negatively correlated at some horizon no longer than $j + 1$:

Proposition 2. If short-run return autocorrelations are positive, then for some $i \leq j$, one must have

$$\text{Cov}(\Delta P_{t+i}, \Delta P_{t-1}) < 0.$$

Interpretation.

- The strategy “buy after any price increase and hold for j periods” must earn zero expected profit when momentum traders are risk-neutral.
- Therefore the initial positive autocorrelation phase must eventually be offset by a negative autocorrelation phase.
- This does *not* contradict Proposition 1. Proposition 1 is conditional on a known news shock. Proposition 2 is an unconditional statement about returns following any price increase, including those that occur late in the momentum cycle.

3.8 Why overreaction happens: the momentum externality

The paper’s most important intuition is the distinction between **early** and **late** momentum trading.

- If a trader buys immediately after a genuine news shock, he earns money because the asset is still underreacting.
- But this purchase itself pushes the price up, making the next generation of traders more optimistic.
- Later traders may then buy at prices already above long-run value.

Interpretation. Early trend followers impose a negative externality on later trend followers. This is why individually sensible momentum trading can generate collectively excessive price pressure.

3.9 Numerical comparative statics in the baseline

The paper performs numerical exercises varying three important parameters.

(i) **Momentum horizon j .** With $z = 12$, $\sigma_\varepsilon = 0.5$, and $\gamma = 1/3$, the peak impulse response is:

- about **1.265** when $j = 6$,
- about **1.342** when $j = 12$,
- about **1.252** when $j = 18$.

This effect is **nonmonotonic**: a longer horizon means more overlapping momentum positions, but also more fear of being caught late in the cycle, so each trader acts less aggressively.

(ii) **Risk tolerance γ .** Higher risk tolerance leads to more aggressive momentum trading, bigger overshooting, and more volatile returns. The effect on pricing error relative to rational expectations is **U-shaped**: some momentum initially helps prices move toward fundamentals faster, but too much creates larger overreaction.

(iii) **Information diffusion speed z .** Slower information diffusion (higher z) makes momentum strategies more profitable, increases ϕ , and tends to produce stronger overshooting and stronger later reversals.

3.10 Extension 1: contrarian trading

The paper next allows a fraction w of arbitrageurs to follow momentum strategies and the rest $(1 - w)$ to follow contrarian strategies based on older lags of returns. Prices become

$$P_t = D_t + \frac{(z-1)\varepsilon_{t+1} + (z-2)\varepsilon_{t+2} + \cdots + \varepsilon_{t+z-1}}{z} + \sum_i (\phi_M \Delta P_{t-i} + \phi_C \Delta P_{t-c-i}).$$

The core equilibrium condition is:

Proposition 3. In an interior solution with both strategies active, the profit opportunities of momentum and contrarian strategies must be equalized.

Interpretation.

- Allowing contrarians cannot eliminate the need for eventual overreaction if there is underreaction in the short run.
- In the numerical examples, contrarians often do not enter at all unless arbitrage capital is very large.
- Even when contrarians are active, they attenuate rather than overturn the hump-shaped response.

3.11 Extension 2: more sophisticated arbitrage and smart money

The authors then consider two richer forms of arbitrage:

- arbitrageurs who can run **bivariate regressions** on both momentum and contrarian signals,
- fully rational **smart-money** traders who observe all signals and the full history of prices.

These smarter traders reduce overshooting, but with *finite* risk tolerance they do not remove it completely.

Proposition 4. With finite-risk-tolerance smart money, any covariance-stationary equilibrium still has:

1. underreaction at impact,
2. active momentum trading,
3. eventual overreaction.

Only if smart money becomes effectively **risk-neutral** do prices follow a random walk and momentum trading disappear.

Interpretation. Fully rational arbitrage attenuates but generally does not eliminate anomalies when arbitrage capital is finite and risky.

4 Identification and instruments (what makes the estimates credible)

4.1 What standard approaches miss

- **Pure rational-expectations models** do not easily generate both short-run continuation and long-run reversal from the same primitive source.
- **Representative-agent behavioral models** of the era can produce underreaction and overreaction, but they do not naturally emphasize how the interaction of heterogeneous trader types generates distinctive cross-sectional and horizon-based predictions.
- **Positive-feedback models** like DeLong et al. are related, but in those models short-run positive autocorrelation often comes from initial overreaction rather than from genuine underreaction followed by momentum-induced overshooting.

4.2 What identifies the mechanism here

The paper's mechanism is pinned down by three assumptions working together:

1. private information diffuses gradually across traders,
2. newswatchers do not extract information from prices,
3. momentum traders use simple strategies and therefore cannot tell whether they are early or late in the momentum cycle.

Interpretation. The last assumption is crucial. If momentum traders could fully condition on all public information, they would know when a price increase reflects fresh news and when it merely reflects prior momentum trades. The momentum externality would then be much weaker.

4.3 Why the auxiliary implications are useful

The model is not just designed to match first-order return autocorrelations. It also delivers sharper predictions that distinguish it from alternatives.

- **Cross section:** Momentum and reversal should be stronger where information diffusion is slower.
- **Public vs private news:** Private-news shocks naturally generate hump-shaped responses; public-news shocks need not.
- **Horizon link:** The timing of reversals should depend on the horizon of the arbitrageurs exploiting momentum.

These are the paper's closest analog to "identification" in an empirical paper: the theory is credible insofar as it generates patterns that competing theories do not easily produce.

4.4 Internal robustness of the mechanism

- **Contrarians.** Allowing contrarian traders does not overturn the core result; at most it dampens the extent of overshooting.

- **Smart money.** Adding fully rational arbitrageurs with finite risk tolerance still leaves underreaction and overreaction in equilibrium.
- **Near-rationality of momentum traders.** The paper emphasizes that momentum traders are not irrational fools; they are nearly rational investors using simple forecasting rules, and can earn positive expected returns.

4.5 Relation to the literature

- **Versus Barberis, Shleifer, and Vishny (1998) and Daniel, Hirshleifer, and Subrahmanyam (1998):** Hong and Stein shift the focus from representative-agent biases to **heterogeneous-agent interaction**.
- **Versus DeLong et al. (1990):** Hong and Stein derive overreaction from underreaction and the dynamics of arbitrage, rather than from an initial overreaction that is subsequently amplified.
- **Broader contribution:** The paper shows how informational externalities among traders can create instability even when each trader is individually optimizing given a limited information-processing rule.

5 Tables and figures

5.1 Figure 1: momentum-trader horizon and the cumulative impulse response

Read:

- The figure varies the momentum traders' holding horizon j while keeping the information-diffusion horizon fixed at $z = 12$.
- The baseline case without momentum trading is monotone: prices drift up toward fundamentals.
- With momentum trading, the response becomes **hump-shaped**.
- The peak overshoot is around **1.265** for $j = 6$, **1.342** for $j = 12$, and **1.252** for $j = 18$.

Main takeaway. A longer trader horizon does not mechanically mean more overreaction. There are two opposing forces: more overlapping generations of momentum traders strengthen propagation, but longer horizons also make traders more cautious because they fear being caught late in the cycle.

5.2 Figure 2: risk tolerance and the cumulative impulse response

Read:

- The figure varies the aggregate risk tolerance γ of momentum traders.
- Higher γ means more aggressive trend-chasing.
- The impulse response becomes more hump-shaped and peaks higher as risk tolerance rises.

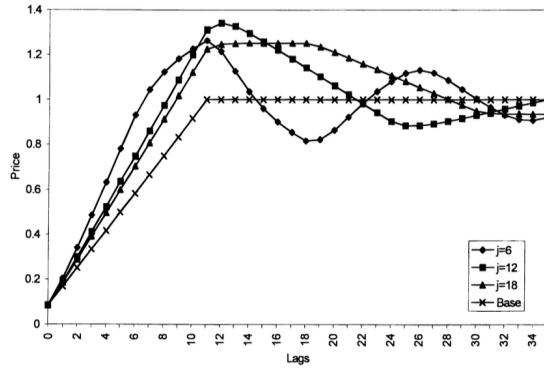


Figure 1. Cumulative impulse response and momentum traders' horizon. The momentum traders' horizon j takes on values of 6, 12, and 18. Base is the cumulative impulse response without momentum trading. The other parameter values are set as follows: the information diffusion parameter z is 12, the volatility of news shocks is 0.5, and the risk tolerance γ is $1/3$.

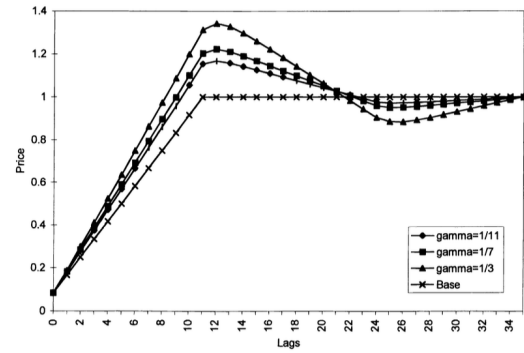


Figure 2. Cumulative impulse response and momentum traders' risk tolerance. The momentum traders' risk tolerance γ takes on values of $1/11$, $1/7$, and $1/3$. Base is the cumulative impulse response without momentum trading. The other parameter values are set as follows: The information diffusion parameter z is 12, the momentum traders' horizon j is 12, and the volatility of news shocks is 0.5.

Main takeaway. More arbitrage capital is not always stabilizing. It can make prices move toward fundamentals faster at first, but it can also produce bigger overshooting and more volatility.

5.3 Figure 3: slower information diffusion means stronger momentum and reversal

Read:

- The figure varies the diffusion parameter z .
- Higher z means slower information flow.
- As z rises, momentum becomes more profitable, momentum traders respond more strongly, and the cumulative impulse response shows stronger overshooting.

Main takeaway. Slower information diffusion strengthens both short-run continuation and long-run reversal. This is the basis for the paper's cross-sectional prediction that small and low-analyst-coverage stocks should display both stronger momentum and stronger eventual reversals.

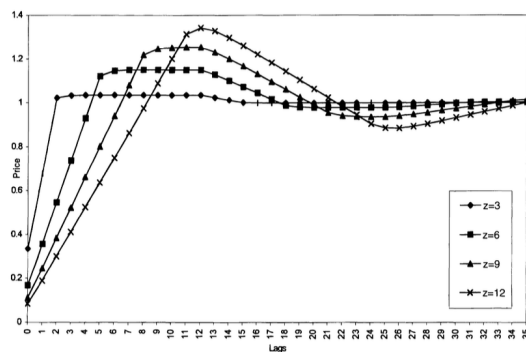


Figure 3. Cumulative impulse response and the information diffusion parameter. The information diffusion parameter z takes on values of 3, 6, 9, and 12. The other parameter values are set as follows: Momentum traders' horizon j is 12, the volatility of news shocks is 0.5, and momentum traders' risk tolerance γ is $1/3$.

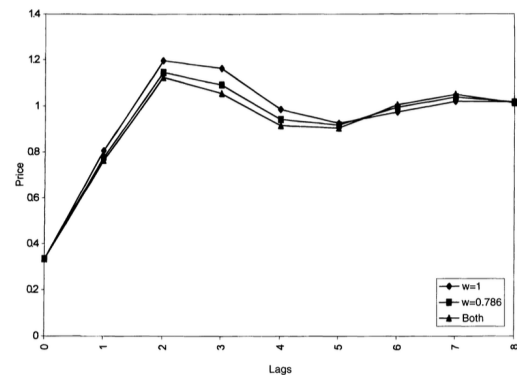


Figure 4. Cumulative impulse response and contrarian trading. Cumulative impulse responses for all-momentum trading equilibrium ($w = 1$); the equilibrium in which traders endogenously choose whether to follow either momentum or contrarian strategies ($w = 0.786$); and the equilibrium in which traders can optimally condition on both momentum and contrarian variables ("Both"). The other parameter values are set as follows: the information diffusion parameter z is 3, the momentum traders' horizon j is 1, the volatility of news shocks is 1, and the risk tolerance γ is $1/0.3$. The contrarians are assumed to trade based on returns from three periods ago.

5.4 Figure 4: contrarian trading and more sophisticated arbitrage

Read:

- The figure compares three cases:
 - all arbitrageurs follow momentum ($w = 1$),
 - an interior equilibrium with both momentum and contrarian strategies ($w = 0.786$),
 - arbitrageurs who can condition on both signals (“Both”).
- The peak response falls as arbitrage becomes more sophisticated, but remains above one.

Main takeaway. More sophisticated arbitrage is stabilizing, but the paper is not knife-edge. The hump-shaped response survives unless arbitrage becomes effectively fully rational and infinitely risk-tolerant.

5.5 Appendix Table A1: comparative statics with respect to horizon

Table A1
Comparative Statics with Respect to Momentum Traders' Horizon
Momentum traders' horizon j takes on values 3, 6, 9, 12, 15, and 18. δ is the intensity of momentum trade described in equation (7). P_t is the stock price at time t . P_t^e is the rational expectations stock price in equation (2). The other parameter values are set as follows: The information diffusion parameter $z = 12$, the volatility of news shocks $\sigma = 0.5$, and the risk tolerance $\gamma = 1/3$.

Momentum Traders' Horizon	$j = 3$	$j = 6$	$j = 9$	$j = 12$	$j = 15$	$j = 18$
δ	0.5550	0.4455	0.3262	0.2065	0.2015	0.2015
Standard deviation of $(P_t - P_t^e)$	0.2239	0.2022	0.2179	0.2029	0.1998	0.1932
Standard deviation of $(P_t - P_t^e)$ at $\phi = 0$	0.9373	0.9373	0.9373	0.9373	0.9373	0.9373
Standard deviation of $(P_t - P_t^e)$	0.8011	0.8438	0.9103	0.9365	0.9524	0.9604
Cumulative impulse response at lag						
0	0.0833	0.0833	0.0833	0.0833	0.0833	0.0833
1	0.2129	0.2038	0.1939	0.1844	0.1855	0.1855
2	0.3682	0.3468	0.3132	0.2801	0.2820	0.2870
3	0.5377	0.4851	0.4355	0.4113	0.3994	0.3912
4	0.6658	0.6028	0.5548	0.5238	0.5071	0.4955
5	0.7530	0.7819	0.8223	0.8365	0.8147	0.7999
6	0.7989	0.8017	0.8059	0.7492	0.7225	0.7042
7	0.8105	0.7946	0.7696	0.7816	0.8302	0.8086
8	0.8287	0.7946	0.7696	0.7816	0.8302	0.8086
9	0.8532	0.7946	0.7696	0.7816	0.8302	0.8086
10	0.8802	0.7946	0.7696	0.7816	0.8302	0.8086
11	0.9030	0.7946	0.7696	0.7816	0.8302	0.8086
12	1.1412	1.2151	1.3389	1.3420	1.2854	1.2471
13	1.1475	1.1263	1.2947	1.3279	1.2909	1.2513
14	1.1040	1.0864	1.3401	1.2969	1.2921	1.2522
15	1.0118	0.9607	1.1820	1.2600	1.2924	1.2523
16	0.9281	0.9012	1.1227	1.2211	1.2736	1.2524
17	0.8782	0.8547	1.0630	1.1817	1.2462	1.2524
18	0.8747	0.8173	1.0032	1.1420	1.2180	1.2524
19	0.9240	0.8228	0.9433	1.1024	1.1848	1.2526
20	0.9977	0.8648	0.8923	1.0627	1.1534	1.2120
21	1.0663	0.9235	0.8500	1.0230	1.1219	1.1664
22	1.1063	0.9834	0.8405	0.9833	1.0904	1.1603
23	1.1012	1.0366	0.8518	0.9436	1.0589	1.1340
24	1.0574	1.0810	0.8733	0.9038	1.0274	1.1076

25	0.9951	1.1175	0.8993	0.8859	0.9929	1.0813
26	0.9382	1.1313	0.9271	0.8848	0.9644	1.0550
27	0.9096	1.1187	0.9537	0.8926	0.9329	1.0286
28	0.9180	1.0870	0.9845	0.9043	0.9022	1.0023
29	0.8572	1.0481	1.0134	0.9175	0.9161	0.9759
30	1.0105	1.0042	1.0395	0.9312	0.9149	0.9496
31	1.0560	0.9658	1.0618	0.9451	0.9146	0.9401
32	1.0786	0.9204	1.0722	0.9590	0.9187	0.9373
33	1.0663	0.9314	1.0719	0.9730	0.9259	0.9365
34	1.0309	0.9076	1.0648	0.9870	0.9344	0.9364
35	0.9961	0.9201	1.0540	1.0010	0.9433	0.9363
Return autocorrelations at lag						
1	0.8630	0.9331	0.9458	0.9466	0.9465	0.9436
2	0.4441	0.7888	0.8415	0.8504	0.8673	0.8834
3	0.1706	0.6063	0.7048	0.7813	0.7388	0.7782
4	(0.1203)	0.4939	0.5561	0.6220	0.6907	0.6920
5	(0.2115)	0.3868	0.4031	0.5056	0.5870	0.6054
6	(0.1035)	(0.0280)	0.2492	0.3846	0.4774	0.5182
7	0.1284	(0.2138)	0.0963	0.2854	0.3658	0.4285
8	0.3635	(0.3253)	(0.0522)	0.1464	0.2537	0.2370
9	0.4735	(0.3607)	(0.1883)	0.0276	0.1415	0.2208
10	0.4086	(0.3533)	(0.3027)	(0.0888)	0.0300	0.1137
11	0.1810	(0.3051)	(0.2708)	(0.2027)	(0.0784)	0.0087
12	(0.0655)	(0.2200)	(0.2955)	(0.2584)	(0.1741)	(0.0840)
13	(0.2180)	(0.0850)	(0.3589)	(0.3385)	(0.2182)	(0.1215)
14	(0.4026)	0.0574	(0.2985)	(0.3346)	(0.2452)	(0.1465)
15	(0.2294)	0.1896	(0.2289)	(0.3088)	(0.2697)	(0.1690)
16	(0.1297)	0.2376	(0.1540)	(0.2739)	(0.2872)	(0.1909)
17	0.1034	0.3632	(0.0823)	(0.2332)	(0.2792)	(0.2125)
18	0.2808	0.2532	(0.0096)	(0.1919)	(0.2595)	(0.2330)
19	0.2397	0.2108	0.0562	(0.1305)	(0.2555)	(0.2485)
20	0.2599	0.1318	0.1177	(0.1083)	(0.2096)	(0.2402)
21	0.0989	0.0331	0.1194	(0.0683)	(0.1603)	(0.2224)
22	(0.1076)	(0.0608)	0.1810	(0.0245)	(0.1486)	(0.2016)
23	(0.2477)	(0.1329)	0.1762	0.0170	(0.1485)	(0.1801)
24	(0.2517)	(0.1785)	0.0573	0.0573	(0.0828)	(0.1583)
25	(0.2046)	(0.1914)	0.1252	0.0929	(0.0510)	(0.1363)

Read:

- The appendix table summarizes how changing the momentum horizon affects pricing errors, volatility, and related moments.
- It complements Figure 1 by showing numerically that the relation between horizon and mispricing is not monotone.

Main takeaway. The duration of trend-following strategies matters quantitatively for both the amount of overreaction and the timing of reversal, which is why the paper puts so much emphasis on the link between investor horizons and return autocorrelations.

6 Short summary

- **Method.** Build a simple behavioral asset-pricing model with two groups of boundedly rational traders: newswatchers and momentum traders.
- **Core mechanism.** Slowly diffusing private information creates underreaction. Simple trend-following arbitrage then amplifies the drift and generates overreaction because arbitrageurs cannot tell whether they are buying early or late in the momentum cycle.

- **Main theoretical result.** Underreaction and overreaction are two phases of one common process, not two unrelated anomalies.
- **Empirical predictions.** Momentum and reversal should be stronger in stocks with slower information diffusion, such as small and low-analyst-coverage stocks; the timing of reversals should depend on arbitrage horizons; private and public news can have different impulse responses.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper provides a unified explanation for **short-run underreaction** / **momentum** and **long-run overreaction** / **reversal**.
- The explanation does not rely on multiple primitive shocks or on a single representative investor with complex psychology.
- Instead, it arises from the interaction between slow information diffusion and simple trend-following arbitrage.

7.2 Economic intuition

- Suppose good private news arrives.
- Newswatchers react, but too slowly, because not all of them see the information immediately and because they do not learn from prices.
- Momentum traders rationally buy after rising prices because those prices usually signal more good news to come.
- But momentum traders cannot tell whether a rising price reflects fresh information or merely earlier momentum purchases.
- So they sometimes buy after the market has already gone too far.
- This makes the same shock produce first underreaction, then overreaction.

7.3 Takeaway

This paper's central lesson is that **boundedly rational arbitrage need not eliminate anomalies; it can transform one anomaly into another**. In the Hong–Stein world, slow diffusion of information creates momentum, and the attempt to exploit momentum generates reversal. That is why the model remains influential: it explains a great deal with a very small set of assumptions, and it generates testable predictions that go beyond simply saying that returns are predictable.